



Information quality, work-family conflict, loneliness, and well-being in remote work settings

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ABSTRACT

Teleworking has rapidly grown and become the prevailing work mode. Married teleworkers may experience increased work-family conflict, while single teleworkers may experience loneliness. This study aims to investigate how teleworkers' information reception quality (Time 1) influences their work-family conflict (Time 2), loneliness (Time 2), and well-being (Time 3) from the perspective of the job demands-resources model. The data were collected through a three-point survey involving 462 participants working from home in Taiwan. The results indicate a negative association between information accuracy and work-family conflict. Work-family conflict, in turn, mediates the relationship between information accuracy and well-being. Information timeliness, as the moderator, weakens the connection between information accuracy and work-family conflict. Additionally, information timeliness is negatively related to loneliness. Loneliness mediates the relationship between information timeliness and well-being. Information accuracy, as the moderator, strengthens the association between information timeliness and loneliness. No impact of information on family-work conflict was observed. Our findings suggest that organizations that convey precise and punctual messages to employees have distinct routes for reducing work-family interference and loneliness and ultimately improving employees' well-being in remote work contexts. This study contributes to the wider telework literature through information experience.

Introduction

Driven by advanced communication technology, teleworking (i.e., remote work) has been growing rapidly and has become a mainstream work mode (Countouris & De Stefano, 2023). It is estimated that 48% of knowledge workers worldwide will work either fully remotely or in a hybrid setting (i.e., work in office at least one day per week) and that virtual workspaces will enhance the capacity to recruit and unite employees irrespective of their geographical location (Gartner, 2023). The main benefits of telecommuting for both companies and professional-level employees are that teleworkers are responsible for their output quality (Felstead et al., 2002; Rodríguez-Modroño & López-Igual, 2021), benefit from flexibility and autonomy (Gajendran & Harrison, 2007; Morgan, 2004; Pérez et al., 2002), experience less negative emotion from the workplace, reduce their stress from commuting (Giménez-Nadal et al., 2020; Pérez et al., 2002), and report being more efficient and having better work-life harmony (McLellan,

2023; Rau & Hyland, 2002).

However, involuntary teleworking following the onset of the pandemic led to challenges and intensified work-family conflict and loneliness (Carnevale & Hatak, 2020) due to the wide-ranging halt in schools and childcare centers and the implementation of social distancing and remote working. Teleworkers working from home (WFH) while married with children may face significant work-family conflict due to the blurring of boundaries between the household and workplace (Clark, 2000; Golden et al., 2006; Lapierre et al., 2016; Leung & Zhang, 2017). On the other hand, single teleworkers without children (Carnevale & Hatak, 2020) and those put in uncertain situations (Doberstein & Charbonneau, 2022; Jämsen et al., 2022) may experience a certain degree of loneliness due to feelings of isolation. The resulting negative pressure and emotions from work-family conflict and loneliness are intense and can have adverse effects on teleworkers' well-being (Kinunen et al., 2006) and affect the organization and individual productivity and performance (DiMaria et al., 2020).

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Carnevale and Hatak (2020) provide insight into whether additional types of organizational support can help combat work-family conflict and loneliness in remote working settings to sustain teleworkers' well-being, and they particularly emphasize "informational support". Similarly, Miyake et al. (2022) suggest that the utilization of information and communication technology by remote workers to interact with supervisors and coworkers can effectively mitigate feelings of loneliness and minimize potential mental health risks. Building on this premise, we intend to investigate the impacts of the quality of information that teleworkers receive on work-family conflict, loneliness, and overall well-being.

Given the nature of teleworking as a result of organizational decentralization (Nilles, 1975), remote workers often encounter decisions regarding task completion and collective responsibility for outcomes, and information quality in terms of its accuracy and timeliness (Forslund, 2007) is crucial for effective decision-making (Andreadakis & Levis, 1987; Ballou & Pazer, 1995). Although information communication technology, e.g., instant messages, online meetings, and internet platforms, serves as an efficient means of communication for organizations and teleworkers, it can also inadvertently lead to miscommunication (unintentional or ineffective) due to the lack of nonverbal cues (such as tone or intonation) in the messages exchanged, increasing the likelihood of misinterpretation (Ramirez et al., 2002; Sarigianni et al., 2017), or communication fluency interrupted by a poor signal and intermittent transmission. Jämsen et al. (2022) examined Finnish public sector employees and discovered that 86% of the participants reported encountering challenges with relational communication when they were shifting to a remote context. The drawback of information reception for teleworkers is that it amplifies stress spillover into their household, causing work-family interference or intensifying loneliness and affecting well-being. Hence, we are interested in investigating the impact of information accuracy and timeliness on the targeted variables.

To explore the hypothesized relationships, the job demands-resources model (JD-R model; Demerouti et al., 2001) was adopted. The JD-R model suggests that employees encounter various factors within the work environment that can be categorized as either demands or resources. Notably, Day et al. (2010) underscored the dual nature of information and communication technology as both a resource and a demand, influencing employees' stress levels and overall well-being. On the positive side, it provides benefits such as improved access to information and increased efficiency. On the negative side, it can create expectations for higher productivity, potentially increasing employees' stress levels. In essence, mitigating work-family conflicts and feelings of isolation among remote workers necessitates organizational attention to effective communication (in terms of accuracy and timeliness). This theoretical perspective helps develop the arguments of this paper.

Overall, this study aims to contribute to the literature on work-family conflict and loneliness among remote workers by investigating the impact of work information timeliness and accuracy. The findings of this research could enhance our understanding of the work-family conflict and loneliness experienced by teleworkers, particularly from the perspective of teleworkers' information-receiving experiences. By delving into these factors, we hope to provide practical recommendations for organizations to support their teleworkers and create a work environment that fosters well-being and productivity. To gather empirical data, we conducted a three-point survey in the context of severe coronavirus infection spreading in Taiwan. This approach allowed us to collect valuable insights from employees working remotely during these challenging times. Fig. 1 depicts the relationships examined in the current study.

2. Hypothesized relationships

2.1. Information quality as resources and demands

According to the JD-R model (Demerouti et al., 2001), job demands require substantial physical or psychological exertion from employees. These demands encompass elements such as work volume, deadline stress, relationship strain, and various other job factors. On the other hand, job resources assist employees in completing their tasks, mitigate the impact of job demands, and hold the potential to foster self-development and growth (Bakker & Demerouti, 2007). In essence, demands deplete employees' physical and psychological reserves, while resources motivate employees and serve as a buffer against high demands (Bakker & De Vries, 2021).

From a resource perspective, information plays a vital role in social support (House, 1981) and is recognized as a valuable resource that mitigates negative perceptions of strain and positively influences various experiences and daily decision-making (Cohen et al., 2000). It involves the provision of advice, suggestions, and factual knowledge, aiding individuals in overcoming challenges and fostering well-being through social connections (House, 1981). On the other hand, from a demand perspective, efficient access to information can be a double-edged sword, as it has the potential to increase workload. For example, the expectation of remaining connected outside work hours can contribute to higher levels of strain (Day et al., 2010, 2019), leading to an escalation of workload, extended work hours, and blurring of work-life boundaries (Rodríguez-Modroño & López-Igual, 2021).

Specifically, for our variables of interest, accuracy reflects the clear and desired response to a given objective, while timeliness describes the ability to meet the required time for a certain task (Andreadakis & Levis, 1987). Accurate and timely information is crucial for successful

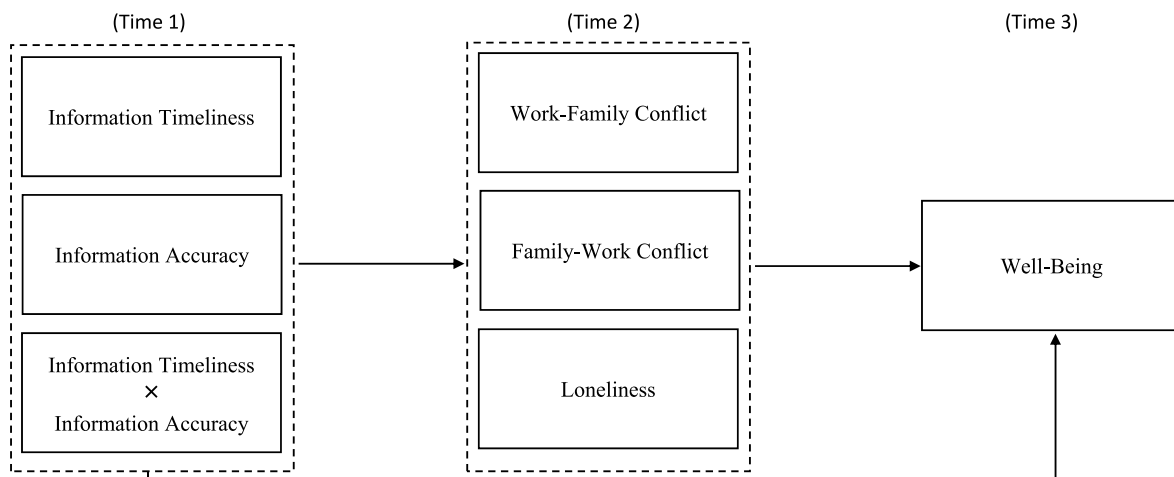


Fig. 1. Research framework.

communication transfer (Forslund, 2007) and serves the purposes of energy savings (i.e., accurate information) and time savings (i.e., timely information). In terms of resources, well-informed employees tend to make superior decisions, facilitate the development and distribution of organizational knowledge (Sharda et al., 1999), and benefit from establishing better leader-member exchange relationships (Sias, 2005). High-quality information determines superior operational performance in collaborative practices involving information sharing, incentive alignment, and joint decision-making (Wiengarten et al., 2010). Additionally, information quality mediates the relationship between information sharing and knowledge creation (Malhotra et al., 2005) and is positively associated with employees' satisfaction when communicating with colleagues (Allen, 1996).

Regarding the demand aspect, timely information may lead to more stress than accurate information. Information communication technology imposes a distinct set of demands in the workplace that necessitate our attention and response (Day et al., 2019; Lowry & Moskos, 2005). These demands encompass social response expectations and cognitive requirements and can serve as work demands and stressors (Barber & Santuzzi, 2015; Day et al., 2012). This situation is associated with an urge to promptly respond to messages upon receiving timely information, and it occurs when work-related information permeates into personal employees' lives (Leung & Zhang, 2017) due to its timeliness. In contrast, precise information enables teleworkers to effectively address work-related matters, take appropriate actions, and reduce uncertainty (Kramer, 1994). This may not directly result in an increased workload because accurate information provides teleworkers with reliable instructions and guidelines, minimizing errors and empowering them to make choices (Andreadakis & Levis, 1987; Ballou & Pazer, 1995). In other words, some aspects of information quality (i.e., timeliness) may serve as both resources and demands for employees, while others (i.e., accuracy) primarily serve as resources.

2.2. Information quality and work-family conflict

Work-family conflict arises when individuals need to balance their responsibilities and requirements between work and family domains (Carlson et al., 2000; Van Daalen et al., 2006). This conflict is bidirectional and involves both work-to-family conflict (WFC) and family-to-work conflict (FWC). It encompasses situations where individuals allocate more time to one domain, leading to insufficient time in the other domain (known as time-based interference), as well as situations where strain experienced in one domain hinders their ability to engage effectively in the other domain (known as strain-based interference). Teleworking can exacerbate work-family conflict through flexible and permeable boundaries between work and personal life (Clark, 2000; Lapierre et al., 2016; Leung & Zhang, 2017). These conflicts can result in job and family discontent, general life strain, significant depressive symptoms, and cognitive exhaustion among individuals (Ford et al., 2007; Frone et al., 1992).

According to the JD-R model, information quality can influence teleworkers' work-family conflicts through both resources and demands. The timeliness and accuracy of information act as resources that reduce uncertainty and cognitive load for teleworkers, enabling them to effectively address work-related matters (Kramer, 1994). As mentioned above, accurate information is an energy-saving resource that equips teleworkers with valuable insights and guidelines, reduces errors and facilitates decision-making (Andreadakis & Levis, 1987; Ballou & Pazer, 1995) to address work tasks. Timely information is a time-saving resource that fosters closer communication and enables positive adjustments to prioritize tasks. Especially in involuntary teleworking, timely information benefits uncertainty reduction due to disconnection from familiar working conditions (Doberstein & Charbonneau, 2022). The delivery of information emphasizes the modification of comprehension over time and produces a mutually reciprocal understanding, which facilitates shared meaning and understanding (Robbins & Judge,

2022, p. 427). Thus, information accuracy and timeliness provide an advantage to teleworkers in reducing work-family conflict.

However, information timeliness may serve as demand and have a negative impact on work-family conflict. For instance, when teleworkers receive instant messages from supervisors or colleagues, their attention can be diverted from the family toward work-related tasks. This timely access to information may contribute to the permeability of the home domain, leading to "technostress" (Leung & Zhang, 2017; Tarafdar et al., 2007). As a result, work-related demands can negatively affect family time and increase workloads, further exacerbating work-family conflict. Building upon the preceding discussion, we propose the following conflicting hypotheses.

Hypothesis 1. Work information quality, in terms of (a) timeliness and (b) accuracy in communications received by teleworkers, negatively predicts work-family conflict.

Hypothesis 2. Information timeliness in communications received by teleworkers positively predicts work-family conflict.

Furthermore, it is noteworthy that information originates primarily from work-related aspects, which may not have an equal impact on WFC and FWC. Research has shown that WFC and FWC are influenced differently by work support and family support (French et al., 2018; Van Daalen et al., 2006). WFC tends to be more influenced by support received from the work domain (i.e., organization, supervisor, and coworker), whereas FWC is more influenced by support from the family domain (i.e., spouse, general family member). Consequently, we anticipate that work information will have a greater influence on WFC than on FWC.

Hypothesis 3. Work information quality, in terms of (a) timeliness and (b) accuracy, has a greater influence on WFC than on FWC.

2.3. Information quality and loneliness

Loneliness is typically defined as a misalignment between an individual's desired and actual levels of social interaction in society (Russell et al., 1980). This negative experience influences physical and psychological health and is strongly associated with stress, anxiety, and distress (Owczarek et al., 2022), and it can increase an individual's risk of stroke, coronary heart disease, and mortality (Holt-Lunstad et al., 2015; Nicole et al., 2016). Teleworkers, especially those who are single and without children, may experience increased feelings of loneliness and social exclusion resulting from a perceived lack of familial connections, a sense of purposelessness, and isolation (Carnevale & Hatak, 2020). Specifically, the abrupt transition to telework during the pandemic posed a significant risk of increased alienation from work as conventional aspects of office life disappear (Rodríguez-Modroño & López-Igual, 2021). There is a sense of detachment from the traditional workplace, which can lead to potential disconnection from familiar working conditions (Doberstein & Charbonneau, 2022). Several surveys have revealed that the sudden shift to remote work has increased employees' stress levels (e.g., Jämsen et al., 2022; Rodríguez-Modroño & López-Igual, 2021). Mass telework significantly contributes to feelings of loneliness due to reduced face-to-face interactions with colleagues and supervisors (Gajendran & Harrison, 2007), especially for individuals with neuroticism in the private sector (Doberstein & Charbonneau, 2022). This lack of social connection has led some employees to express a desire to return to the office, citing feelings of isolation, loneliness, and depression associated with working from home (Bloom et al., 2015).

According to the JD-R model, information timeliness and accuracy can serve as resources (Day et al., 2010) for diverting workers' attention from psychological feelings of loneliness by providing them with real-time updates and communication that create a sense of connectedness and engagement. Importantly, we did not view timeliness information as a demand for teleworkers experiencing loneliness. Rather, instant messaging can serve as a means for lonely teleworkers to connect

with others and draw their attention from loneliness. Addressing loneliness involves establishing, revitalizing, and maintaining social identities through significant associations within groups (Haslam et al., 2022), and timely information plays a central role in facilitating these connections. Instant messaging platforms or email formats enable teleworkers to stay connected with their colleagues and team (Zaccaro & Bader, 2003), allowing them to engage in spontaneous conversations and share updates, which can help alleviate feelings of loneliness (Miyake et al., 2022). Regarding information accuracy, feedback and performance evaluations from supervisors or managers help teleworkers stay connected to their work and receive recognition, reinforcing their sense of value, fostering a sense of security in their work responsibilities and boosting their efficacy in completing work assignments (Dewett & Jones, 2001). Expanding on the previous discourse, we present the following research hypothesis.

Hypothesis 4. The quality of work information, in terms of (a) timeliness and (b) accuracy, received by teleworkers negatively predicts loneliness.

2.4. Information quality and well-being

Well-being is an individual's state of having positive emotions (such as happiness), the absence of adverse emotions (such as anxiety), and an overall feeling of contentment with life (Diener, 1984), and the broad meaning of well-being encompasses both "effective functioning" and "feeling good" (Kowalski & Loretto, 2017). The significance of well-being is gaining interest among practitioners and researchers due to its critical role in the sustainability of organizations (Barrett, 2017).

Although creating the conditions for well-being and productivity in a remote or hybrid setting can initially seem difficult (Fosslien & Cohen, 2023), ensuring information quality delivery with accuracy and timeliness as a resource (Day et al., 2010) may reduce work-related uncertainty for teleworkers, increase teleworkers' sense of control in accomplishing the task (Teo et al., 1998), and improve their problem-solving capacity through information accessibility (Morgan, 2004). Studies have demonstrated that increased employee information is associated with reduced job tension, improved work effectiveness, and heightened psychological empowerment (Laschinger et al., 2001). These favorable attributes have the potential to significantly contribute to the well-being of teleworkers.

Hypothesis 5. The quality of work information, in terms of (a) timeliness and (b) accuracy, received by teleworkers positively predicts well-being.

2.5. Interactions between timeliness and accuracy

In addition to the direct effects of information quality on the target variables (i.e., work-family conflict, loneliness, and well-being), information accuracy may moderate the relationship between information timeliness and the target variables. Conversely, information timeliness may moderate the association between information accuracy and target variables. Specifically, information timeliness may function as both a resource and a demand for teleworkers, while information accuracy primarily serves as a resource. Therefore, investigating the interaction effects between information timeliness and accuracy can provide deeper insight into understanding the roles of information timeliness and accuracy as resources and demands in the context of teleworkers' work-family conflict, loneliness, and well-being.

Hypothesis 6. Information timeliness moderates the relationships between information accuracy and (a) work-family conflict, (b) loneliness, and (c) well-being.

Hypothesis 7. Information accuracy moderates the relationships between information timeliness and (a) work-family conflict, (b) loneliness, and (c) well-being.

2.6. Work-family conflict and loneliness as mediators between information quality and teleworkers' well-being

In the context of teleworking, work-family conflict and loneliness have become notable health crises (Carnevale & Hatak, 2020) that are more intense than those in the workplace. Numerous studies have shown the negative impact of work-family conflict on individuals' well-being due to high levels of psychological strain transmitted directly and indirectly (e.g., Amstad et al., 2011; Grant-Vallone and Donaldson, 2001; Lunau et al., 2014). Loneliness has been shown to longitudinally predict subjective well-being (Saygin et al., 2015; VanderWeele et al., 2012) and has substantial psychological and health impacts, hence influencing the well-being of teleworkers. Taken together, the inferences above provide accurate and timely information related to reducing teleworkers' work-family conflicts and loneliness, which in turn is associated with improved well-being. Thus, work-family conflict and loneliness mediate the relationship between information timeliness and the well-being of teleworkers.

Hypothesis 8. (a) Work-family conflict and (b) loneliness mediate the relationship between information timeliness and teleworker well-being.

Hypothesis 9. (a) Work-family conflict and (b) loneliness mediate the relationship between information accuracy and teleworker well-being.

3. Method

3.1. Respondents and procedure

Data were gathered from a broader data collection effort for different research purposes during mid-2021, coinciding with the Taiwanese government's remote work advocacy to restrict the spread of the coronavirus (Chuang et al., 2023). The onset of the coronavirus led to the global embrace of telework systems. International companies are choosing to recruit teleworkers to sustain operations, and Taiwan is ranked among the top regions in Asia for talent competitiveness (IMD, 2023), making it a favored location. Utilizing a Taiwanese sample can enhance our understanding of remote workers in the Asian and Chinese cultural context.

A market research company was assigned the task of selecting and contacting work-from-home employees from listed companies and public universities in Taiwan. The firm was authorized to randomly select samples from this specified pool, with the requirement of securing at least 300 teleworker participants. The three-wave survey was administered over an interval of a week. Adopting a temporal design is beneficial for examining the theoretical model of change and considering the use of time to better understand psychological processes (Wright & Markon, 2016). Participants were sent a unique link via mobile message and e-mail to access the questionnaire. The first survey included an informed consent form and collected information on timeliness, accuracy, and demographic variables. The second survey included questions about work-family interference and loneliness. In the final phase of the survey, participants' subjective well-being was measured.

Initially, 517 participants consented to complete the survey, and 462 participants (89% response rate) completed all three questionnaires. The occupations/professions of the majority of participants were as follows: 62.1% were administrative employees (including human resources/finance/administration), 9.3% were in R&D (research and development/technology), 7.6% were in marketing/sales, 5.8% were engaged in production-related work (manufacturing/engineering/quality control), and 15% were others. Of the 462 participants, the average number of work-from-home workdays per week was 1.90 ($SD = 1.79$), with 11.9% working from home five days a week, 8.4% working four days a week, 19% working three days a week, and others working two or less days a week; 74% were female; the average age was 38.77 years ($SD = 8.61$); the average tenure was 9.33 years ($SD = 8.19$); 66.7% had a bachelor's

degree, and 27.4% had a graduate degree; 54.5% were married; and 46.3% had children.

3.2. Measures

Information timeliness and information accuracy were measured with 7 items (4 for timeliness and 3 for accuracy) revised from Lan and Sie's (2010) capabilities of media richness scale. The original scales were measured in the education domain, and we revised the scales to be suitable for the work domain. Items were scored on a 5-point rating scale (1 = *strongly disagree*; 5 = *strongly agree*). Sample items are "When the company posts an announcement, I can receive the message immediately" and "I can easily understand the messages I receive from my company". Cronbach's α for information timeliness was 0.80, and that for information accuracy was 0.88.

Work-family conflict was measured by 9 items (4 for WFC and 5 for FWC) developed by Carlson et al. (2000), including time-based and strain-based situations. Items were scored on a 5-point rating scale (1 = *strongly disagree*; 5 = *strongly agree*). Sample items are "The time I must devote to my job keeps me from participating equally in household responsibilities and activities" and "The time I spend on family responsibilities often interferes with my work responsibilities." Cronbach's α was 0.91 for WFC and 0.87 for FWC. To validate the distinct structures of WFC and FWC, a confirmatory factor analysis (CFA) was conducted with a two-factor model in which WFC is composed of 4 items and FWC is composed of 5 items, $\chi^2(24) = 56.28$, $CFI = 0.99$, $TLI = 0.98$, $RMSEA = 0.03$, $SRMR = 0.05$.

Loneliness was measured using Russell et al.'s (1980) 20-item scale of subjective feelings of loneliness. The items were scored on a 5-point rating scale (1 = *I never feel this way*; 5 = *I always feel this way*). A sample item is "I am unhappy doing so many things alone." Cronbach's α was 0.96.

Well-being was measured with 20 items of the short version of the Chinese Happiness Inventory (Lu & Lin, 1998). Items were scored on a 7-point rating scale (1 = *strongly disagree*; 7 = *strongly agree*). A sample item is "I am well satisfied about everything in my life". Cronbach's α was 0.97.

Control variables

We controlled for age, sex, marital status, and children because these variables have been found to influence work-family conflict (e.g., Giménez-Nadal et al., 2020; Van Daalen et al., 2006; Yuan et al., 2022); additionally, we included remote work days as a control variable to determine whether they influenced the results. In investigating the impact of informational accuracy and timeliness on WFC, we considered the effects of FWC and loneliness as control variables, and vice versa. This approach was adopted to minimize the possible spill-over of generalized negative feelings among the WFC, FWC, and loneliness constructs (Golden et al., 2006).

3.3. Analysis approaches

To analyze our data, we initially examined all variables constituting distinct constructs with a CFA test using Mplus 8.6. For hypothesis testing, we performed hierarchical regression analysis with SPSS 22. Prior to the regression analyses, standardization through centering (Robinson & Schumacker, 2009) was applied to all variables to mitigate multicollinearity. Variance inflation factors (VIFs) were also computed for each regression analysis to assess the extent of multicollinearity among variables (Cohen, 2013).

To examine mediation and moderation effects, we utilized Hayes' (2022) PROCESS Macro for SPSS, a versatile tool for ordinary least squares (OLS) and logistic regression path analysis with preset models. PROCESS supports single- and multiple-mediator models, two- and three-way interactions in moderation models, and moderated mediation models. Additionally, bootstrapping confidence intervals are applied to

draw random samples with replacement from the data, which is particularly beneficial for non-normal distributions and small sample sizes (Hayes, 2022), and the results are largely consistent with those of structural equation modeling (Hayes et al., 2017). In addition, we use the Sobel (1982) test to further confirm the mediating roles of these variables. The Sobel test is divided by its standard error estimate, resulting in a Z-statistic. The Z-statistic can then be compared to a standard normal distribution to assess statistical significance. Specifically, we utilized the PROCESS Macro Model 4 and the Sobel test to examine multiple mediating roles between the independent variables and a specific dependent variable.

4. Results

Before testing the hypothesized relationship, CFAs were adopted to ensure that the variables under investigation were distinct constructs. Parceling was used for six major study variables (*information timeliness, information accuracy, WFC, FWC, loneliness, and well-being*), and the results show that the model fit the data adequately $\chi^2(1465) = 4038.18$, $CFI = 0.88$, $TLI = 0.88$, $RMSEA = 0.06$, $SRMR = 0.05$.¹ The specific cutoff values may vary depending on sample size, the estimators used, and distribution. Kline (2016) suggests that a CFI and TFI higher than 0.85 indicates an acceptable level of fit, while RMSEA and SRMR values less than 0.08 are considered indicative of a good fit (Hu & Bentler, 1999). Table 1 displays the descriptive statistics, including the means, standard deviations, and intercorrelations, of all the variables examined in the study. Initially, information timeliness and accuracy were negatively correlated with WFC, FWC, and loneliness but positively correlated with well-being.

To test the hypotheses, we conducted hierarchical regression analysis and examined the variance inflation factors (VIFs), finding values ranging from 1.03 to 2.77, well below the cutoff of 10 (Cohen, 2013). As shown in Table 2, information accuracy showed a negative relationship with WFC ($\beta = -0.15$, $p < 0.01$). Although information timeliness did not have a significant effect on WFC ($\beta = 0.01$, $p = .ns$), the interaction effect between information accuracy and information timeliness positively affected WFC ($\beta = 0.08$, $p < 0.05$), supporting Hypothesis 1(b) and Hypothesis 6(a).

Information timeliness ($\beta = 0.03$, $p = .ns$) and information accuracy ($\beta = -0.04$, $p = .ns$), and their interaction ($\beta = -0.00$, $p = .ns$) did not have significant effects on FWC, thus supporting Hypothesis 3.

Information timeliness was negatively related to loneliness ($\beta = -0.22$, $p < .01$), although information accuracy did not have a significant effect on loneliness ($\beta = 0.02$, $p = .ns$). The interaction effect between information accuracy and information timeliness negatively affected loneliness ($\beta = -0.09$, $p < .05$), supporting Hypothesis 4(a) and Hypothesis 7(b).

As shown in Table 3, both information timeliness ($\beta = 0.14$, $p < .05$) and information accuracy ($\beta = 0.17$, $p < .05$) had positive significant effects on well-being, and the interaction effect between information accuracy and information timeliness marginally positively affected well-being ($\beta = 0.09$, $p < .10$), supporting Hypotheses 5(a) and 5(b).

In addition, some argue that education and occupation may also influence work-family conflict (e.g., Dierdorff & Ellington, 2008). We conducted additional analyses to control for education and occupation, and these two variables did not significantly influence our results when included (see Table B and C in the Supplemental Analysis).

Figs. 2 and 3 illustrate the interaction effect between low and high levels of information accuracy and information timeliness on work-family conflict and loneliness. The effect of information accuracy on work-family conflict was less pronounced for high-level information timeliness than for low-level information timeliness. Additionally, the

¹ We also calculated each variable CFA results in Table A of Supplement Analysis.

Table 1
Descriptive statistics and correlations.

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10
1 Age	38.77	8.61	–									
2 Gender	.26	.44	–.05	–								
3 Marriage	.55	.50	.47**	–.06	–							
4 Have Children	.46	.50	.51**	–.09	.78**	–						
5 Remote Workdays	1.90	1.79	–.05	–.13**	–.01	–.04	–					
6 Information Timeliness	3.58	.74	.12**	.02	.09	.09	.07	–				
7 Information Accuracy	3.59	.80	.11*	–.01	.07	.07	.10*	.75**	–			
8 Work-Family Conflict	2.46	1.04	–.06	.09*	–.04	–.04	–.02	–.22**	–.28**	–		
9 Family-Work Conflict	2.19	.87	–.04	.12*	.10*	.11*	–.00	–.17**	–.21**	.60**	–	
10 Loneliness	2.04	.69	–.11*	.04	–.14**	–.12**	–.02	–.29**	–.25**	.32**	.39**	–
11 Well-being	4.50	1.11	.13*	–.02	.11*	.12**	.01	.28**	.26**	–.27**	–.26**	–.49**

Note. n = 462 participants. For gender, 1 = male, 0 = female; for marriage, 0 = single, 1 = married; for have children, 0 = no, 1 = yes.

* $p < .05$; ** $p < .01$, two-tailed test.

Table 2
Hierarchical regression results.

	Work-Family Conflict			Family-Work Conflict			Loneliness		
	Model1	Model2	Model3	Model1	Model2	Model3	Model1	Model2	Model3
<i>Step 1. Control variables</i>									
Age	.04	.05	.05	–.10*	–.10*	–.10*	–.01	.01	.01
Gender	.01	.02	.02	.07 ⁺	.07 ⁺	.07 ⁺	–.02	–.01	–.02
Marriage	–.05	–.05	–.05	.10 ⁺	.11 ⁺	.11 ⁺	–.12 ⁺	–.11 ⁺	–.10
Have Children	–.07	–.07	–.06	.13*	.13*	.13*	–.06	–.05	–.06
Remote Workdays	–.02	–.01	–.01	.03	.03	.03	–.03	–.01	–.01
Work-Family Conflict	–	–	–	.53**	.53**	.53**	–	–	–
Family-Work Conflict	.59**	.57**	.56**	–	–	–	.10 ⁺	.07	.08
Loneliness	.08 ⁺	.05	.06	.23**	.23**	.23**	.34**	.33**	.32**
<i>Step 2. Main effect</i>									
Information Timeliness		.01	.02		.03	.03		–.22**	–.23**
Information Accuracy		–.15**	–.13*		–.04	–.04		.02	.01
<i>Step 3. Interaction effect</i>									
Information Timeliness × Information Accuracy			.08*			–.00			–.09*
Overall F	40.30**	33.72**	30.96**	51.99**	40.34**	36.23**	14.83**	14.70**	13.72**
R ²	.38	.40	.41	.45	.45	.45	.19	.23	.23
ΔR ²	.37	.39	.39	.44	.43	.43	.17	.21	.22

Note. n = 462 participants. + $p < .10$; * $p < .05$; ** $p < .01$. The entries are standard regression coefficients.

Table 3
Hierarchical regression results.

	Well-being			
	Model1	Model2	Model3	Model4
<i>Step 1. Control variables</i>				
Age	.08	.05	.05	.05
Gender	–.00	–.01	–.00	.03
Marriage	.04	.03	.02	–.00
Have Children	.05	.05	.06	.05
Remote Workdays	.02	–.01	–.01	.01
<i>Step 2. Main effect</i>				
Information Timeliness		.15*	.16*	–
Information Accuracy		.16*	.17*	–
Work-Family Conflict				–.10 ⁺
Family-Work Conflict				–.03
Loneliness				–.44**
<i>Step 3. Interaction effect</i>				
Information Timeliness × Information Accuracy			.09 ⁺	
Overall F	2.01 ⁺	7.12**	6.70**	20.31*
R ²	.02	.10	.11	.26
ΔR ²	.01	.09	.09	.25

Note. n = 462 participants. + $p < .10$; * $p < .05$; ** $p < .01$. The entries are standard regression coefficients.

impact of information accuracy on loneliness decreased to a greater extent with high-level information accuracy than with low-level information accuracy.

Following the finding of the main effect, only Hypotheses 8 (b) and 9

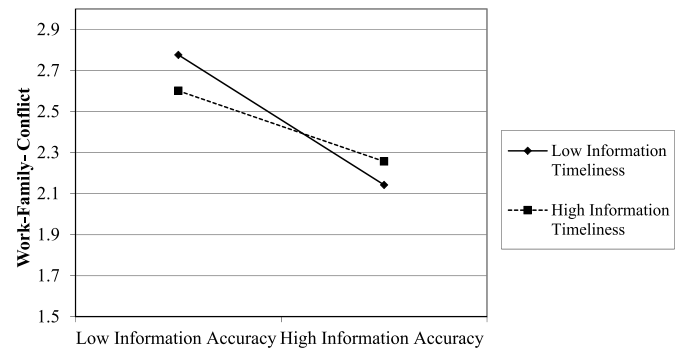


Fig. 2. The moderation of information timeliness on the relationship between information accuracy and work family conflict.

(a) regarding the mediating role of WFC and loneliness were tested; the other paths were not tested because the main effects were not significant. We conducted Hayes' (2022) Process Macro Model 4 with a sample size of 10,000 bootstraps and a Sobel test. The results supported the relationship between informational timeliness and well-being via loneliness, with an indirect effect of 0.19 (95% CIs = 0.11 0.27), which excluded zero. Additionally, the Sobel test ($Z = 5.73$, $SE = 0.04$, $p < .01$) provided further support. Similarly, the relationship between informational accuracy and well-being via WFC was supported, with an indirect effect of 0.08 (95% CIs = 0.03 0.13), excluding zero. The Sobel test ($Z = 4.28$, $SE = 0.02$, $p < .01$) further confirmed the mediating effect (see

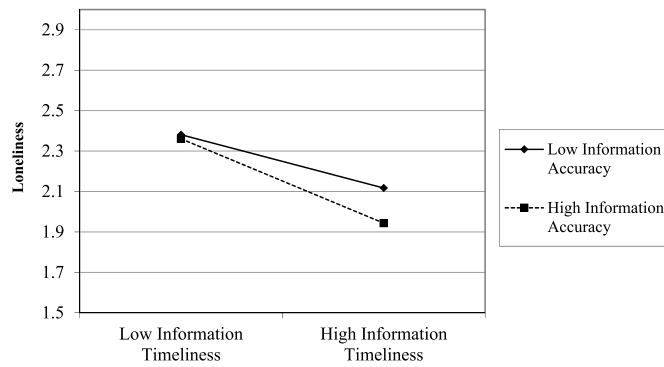


Fig. 3. The moderation of information accuracy on the relationship between information timeliness and loneliness.

Table 4

Mediation results.

Paths and effects	Conditional indirect effect	95% confidence intervals	
		Lower	Upper
<i>Mediation Results</i>			
IA→WFC→Well-being	.08	.03	.13
IT→Lonely→Well-being	.19	.11	.27

Note. n = 462 participants. Bootstrap sample size = 10,000.

IA = information accuracy; IT = information timeliness; WFC = work-family conflict.

Table 4).

5. Discussion

Advanced information technology reshapes organizational work modes through flexible workplaces. Teleworking has accelerated and enabled effective collaboration, especially during the coronavirus epidemic. However, remote work is dual-edged (Lowry & Moskos, 2005) because working from home generates more vital work-family conflict and loneliness (Carnevale & Hatak, 2020), resulting in impaired teleworker well-being (Felstead et al., 2002; Golden et al., 2006). Informational support through communication technology may be useful in mitigating the pressure and negative feelings associated with remote settings (Carnevale & Hatak, 2020; Miyake et al., 2022).

Our findings show that organizational information accuracy negatively affects participants' work-family conflicts, information timeliness negatively affects participants' loneliness, and both information timeliness and information accuracy positively affect the well-being of those participants in remote-work setups. In addition, work-family conflict mediates the relationship between information accuracy and well-being, with information timeliness moderating this association. Loneliness mediates the relationship between information timeliness and well-being, with information accuracy moderating this relation. However, we did not find that information timeliness and information accuracy have effects on family-work conflicts.

These interesting findings suggest that teleworkers' information-receiving can have a more direct impact on work-family conflict, loneliness, and well-being. However, it may not significantly influence family-work conflict. Specifically, information accuracy is found to be beneficial for reducing the work-family conflict experienced by teleworkers. This could be attributed to the precision of the information, which assists employees in understanding how to handle tasks and solve problems effectively. On the other hand, information timeliness is found to be beneficial for reducing teleworkers' feelings of loneliness. Timely information allows them to actively engage with their work team and diminishes their sense of isolation. Consistent with the JD-R model, our findings indicate that information accuracy is an essential resource for

teleworkers to alleviate work-family conflict, while information timeliness is a significant resource for decreasing loneliness.

Additionally, information accuracy moderates the relationship between information timeliness and loneliness. This suggests that when information is accurate, it enhances the effect of information timeliness in reducing teleworkers' feelings of loneliness. On the other hand, the moderating effect of information timeliness on the relationship between information accuracy and work-family conflict reveals a weak negative impact. Timely information might make teleworkers slightly apprehensive, as they strive to balance their time and energy between work and household responsibilities. This suggests that the immediate availability of information may amplify the impact of cognitive workload on teleworkers experiencing work-family conflict. These findings highlight the significance of timely information as a demand (Day et al., 2010; Lowry & Moskos, 2005) that can increase strain among teleworkers, especially those who have family roles. See our results model in Fig. 4.

Overall, this study suggests that organizations actively provide accurate and timely information to help teleworkers decrease uncertainty and address work-family conflict and loneliness, consequently improving their well-being, which affects productivity and effectiveness.

5.1. Theoretical implication

Given the nature of remote work, work-family conflicts and loneliness are prominent concerns, and this study adopts a unique perspective of information quality to investigate these challenges. We address the research gap highlighted by Carnevale and Hatak (2020), who noted that loneliness is less studied than work-family conflict in remote working contexts. Moreover, our study adopts a unique perspective by examining the role of information accuracy and timeliness, an aspect that has received limited attention in previous research. The original idea was that information timeliness and accuracy could alleviate stress by facilitating convenient access to information, eliminating the need for excessive time and travel to obtain necessary information (Day et al., 2010), and playing a crucial role in decision-making processes (Andreidakis & Levis, 1987; Ballou & Pazer, 1995). Drawing upon the JD-R model (Demerouti et al., 2001), our study elucidates the impacts of organizational information accuracy and timeliness on teleworkers' work-family conflict and loneliness.

Our findings suggest that information accuracy and timeliness serve as resources for teleworkers in managing work-family conflict, loneliness, and overall well-being. Information accuracy helps teleworkers alleviate work-family conflict and reduce uncertainty. On the other hand, information timeliness reveals a demand for teleworkers to effectively manage work-family conflict, which may moderate the impact of information accuracy on work-family conflict. Additionally, information timeliness serves as a significant resource for teleworkers in addressing loneliness and reducing uncertainty, facilitating their connection with supervisors and team members in the organization. Moreover, information accuracy strengthens these relationships. Overall, both information accuracy and timeliness contribute to teleworkers' well-being, and work-family conflict and loneliness have different mediating effects on well-being. Notably, loneliness has a stronger impact on teleworkers' well-being.

In addition, because information accuracy and timeliness are work resources, these variables did not have a significant impact on family-work conflict. These results align with previous findings (French et al., 2018; Van Daalen et al., 2006) that emphasize the stronger influence of work support in reducing work-family conflict than in reducing family-work conflict.

5.2. Practical implications

According to the latest trend report, there is a significant demand for remote work opportunities. Fifty percent of employees prioritize remote

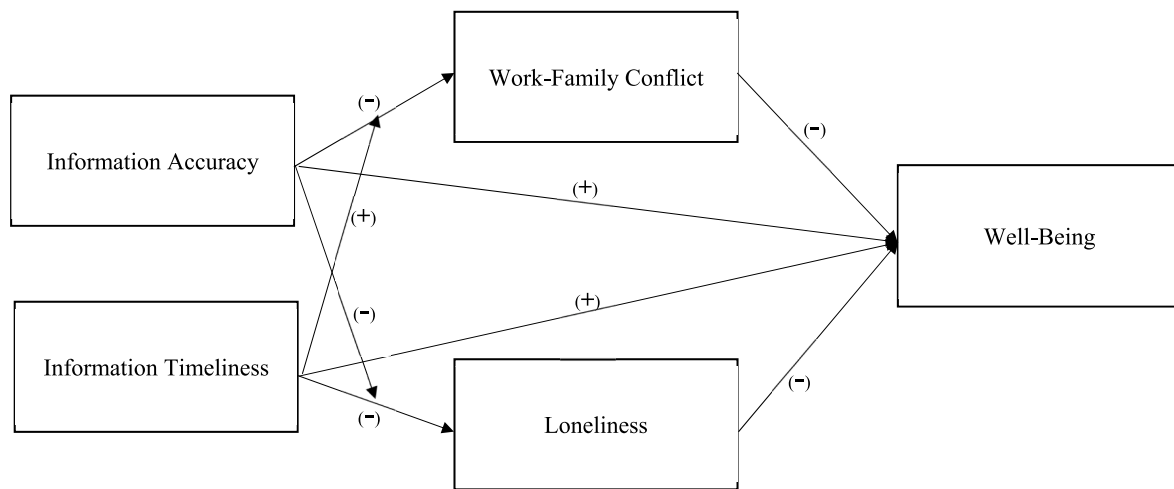


Fig. 4. Result model.

work for their next job, while only 4% prefer fully in-person work (Fosslien & Cohen, 2023). Similarly, approximately half of the respondents from British Columbia expressed that they would contemplate resigning from their current job if their organization did not provide them with the desired level of telework flexibility (Canseco, 2021). However, teleworkers who have family responsibilities experience higher levels of work-family conflict, while single teleworkers often experience greater feelings of loneliness (Carnevale & Hatak, 2020), which influences their well-being.

To address these challenges, leveraging information technology effectively and providing information support to teleworkers should be considered crucial and highly feasible approaches. Our study serves as a reminder to organizations and managers about the importance of prioritizing communication performance, particularly in terms of timeliness and accuracy. According to a 2022 Cisco hybrid work report across 27 countries with 28,025 full-time employees, only 67.9% of respondents reported that their company had addressed technology and connecting issues (McLellan, 2023). This indicates that there is still a gap in remote communication improvement within many organizations. Organizations should account for various factors, including information transmission and the diverse needs of teleworkers, to balance the convenience and user-friendliness needs of hybrid and remote workers.

Notably, both the organizations and teleworkers may face interruptions in information timeliness due to poor signal or intermittent transmission. This highlights the need for technological solutions, such as signal amplifiers and communication jamming, to address these issues and ensure consistent and timely information flow. Additionally, both managers and teleworkers need to be knowledgeable about effective communication strategies, including controlling communication to be accurate, achieving interpersonal exchange and organizational goals, and being perceived as competent communicators (Kramer, 1994).

5.3. Limitations and future directions

We collected responses from employees who worked from home in Taiwan. It is essential to highlight that these individuals were not traditional teleworkers but were instead influenced by infectious disease policies. This highlights the need for further surveys that specifically target typical teleworkers to delve deeper into their experiences and investigate related factors.

In addition, our data are self-rated with time-lagged surveys; although self-reported data may increase the risk of bias (Bailey & Kurland, 2002), self-reported data can also provide insight into participants' perceptions and attitudes, which may not be observable through other means (Baker & Brandon, 1990). Future studies should collect data

from multiple sources, including coworkers, spouses, and supervisors, to minimize prejudice. The data are correlational in nature, which prevents causal explanations from being drawn, and future studies should adopt an experimental design to examine such relationships.

Moreover, our study provides positive findings regarding the role of information accuracy and timeliness in mitigating work-family conflict and loneliness but also emphasizes the perspective of information as a resource rather than a demand in teleworkers' well-being. Indeed, the provision of excessive organizational information can potentially result in information overload and increased workload (Day et al., 2010), and increased use of information communication technology can lead to higher levels of stress, mental fatigue and difficulty concentrating, especially for individuals who are frequently exposed to information communication technology (Barber & Santuzzi, 2015; Day et al., 2012). Subsequent research should explore the potential negative effects of excessive information and communication duration on teleworkers and whether such information causes intrusions and multitasking and the consequences of burnout or mental fatigue.

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CRediT authorship contribution statement

Ya-Ting Chuang: Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Hua-Ling Chiang:** Data curation, Writing – review & editing, Conceptualization. **An-Pan Lin:** Conceptualization, Data curation, Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2024.108149>.

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